

陈信均

Homework for Chapter 4

Calculate the spectra of reflection for a single surface for no damping, weak ($g=0.1 \Delta_{LT}/\hbar$) and strong damping ($g=0.6 \Delta_{LT}/\hbar$). Compare the shift of the reflection maxima and minima with respect to the transverse and longitudinal eigenenergies ω_0 and ω_L , respectively. Which quantity can be deduced with reasonable accuracy from a first inspection of the reflection spectra?

$$\xi(\omega) = \epsilon_0 + \frac{f}{\omega_0^2 - \omega^2 - i g \omega}$$

$$\epsilon_0 + \frac{f}{\omega_0^2 - \omega_L^2 - i g \omega_L} = 0$$

$$\Delta_{LT} = \omega_L - \omega_0, \quad R(\omega) = \left| \frac{n(\omega) - 1}{n(\omega) + 1} \right|^2 = \left| \frac{\sqrt{\xi(\omega)} - 1}{\sqrt{\xi(\omega)} + 1} \right|^2$$

①. $g=0$ 无阻尼

$$\omega_0 < \omega < \omega_L$$

$$\epsilon < 0, \text{ 全反射 } R=1$$

最大值最小值很接近 ω_0, ω_L

②. $g = \frac{0.1 \Delta_{LT}}{\hbar}$ 弱阻尼

最小值处 ω 略大于 ω_L

最大值处 ω 略于 ω_0

③. $g = \frac{0.1 \Delta_{LT}}{\hbar}$ 过阻尼

最大值更加远离 ω_0 , 最小值略大于 ω_L

ω_0 可以被准确估计

